

Electrochemical energy conversion

MT25-30104 - Material Konversi Energi

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Topik Bahasan

- Water splitting
- Fuel Cells

Water Splitting

- Air → molekul sederhana, terdiri dari 2 hydrogen dan 1 oksigen

Apa itu *water splitting*?

Apa saja manfaat air?

Kenapa harus dipecah “*split*” ?

Why to split water?

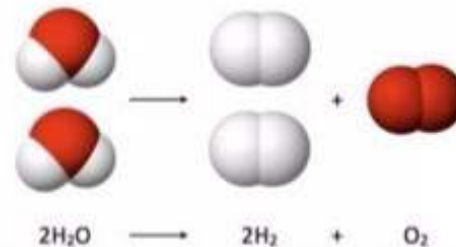
- 70% bumi terdiri dari air (melimpah)
- Produksi bahan bakar hydrogen semakin diperlukan karena minyak dan non-renewable fuel semakin berkurang dan mahal
- Mengurangi emisi gas rumah kaca
- Dapat menjadi komponen kunci *hydrogen economy*

Methods

- Electrolysis
- **Photoelectrochemical Water Splitting**
- Photocatalytic Water Splitting
- Radiolysis
- Photobiological Water Splitting
- Thermal decomposition of Water
- Nuclear thermal
- Chemical production, etc

Electrolysis, Radiolysis, Thermolysis

- If an electrical current is passed through water between electrodes (the positive and minus poles of a battery), the water is split into oxygen and hydrogen.
- Used in industries especially in making aluminium.
- Two types
 - High – pressure electrolysis
 - High – temperature electrolysis



- Radiolysis: **Radiolysis** is the dissociation of molecules by **nuclear radiation**. Here, it is a separation of water via radioactive activity. The end result is the production of hydrogen peroxide.
- Thermolysis: **Thermolysis** is defined as a chemical reaction whereby at **elevated temperatures** water molecules split into their atomic components hydrogen and oxygen.

Photoelectrochemical devices...

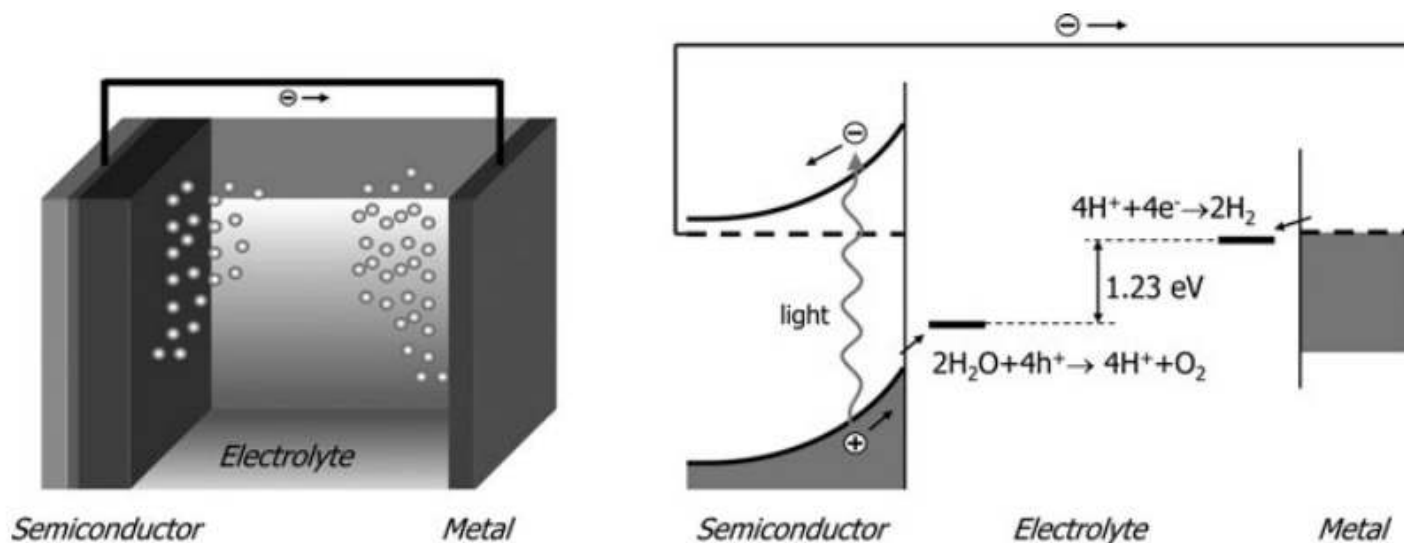
- **Elektrokimia** sebagian besar diterapkan dalam sintesis, analisis kimia dari penyimpanan energi, dan aspek konversi.
- Sel **fotoelektrokimia** adalah perangkat yang memanfaatkan sumber cahaya ke semikonduktor (atau fotosensitizer) untuk menghasilkan energi listrik atau untuk memicu reaksi kimia untuk menyimpan energi dalam bentuk ikatan kimia, contohnya produksi hidrogen dengan *water splitting*.

Photoelectrochemical devices...

- Salah satu perangkat penghasil energi terpenting melibatkan pemisahan fotoelektrokimia air menjadi hidrogen dan oksigen → **hydrogen-producing photoelectrochemical cell (HPPEC)**
- Perangkat lain adalah sel surya fotoelektrokimia di mana reaksi redoks dilakukan di bawah penerangan sinar matahari untuk menghasilkan energi listrik → **electricity-producing photoelectrochemical cells (EPPEC)**

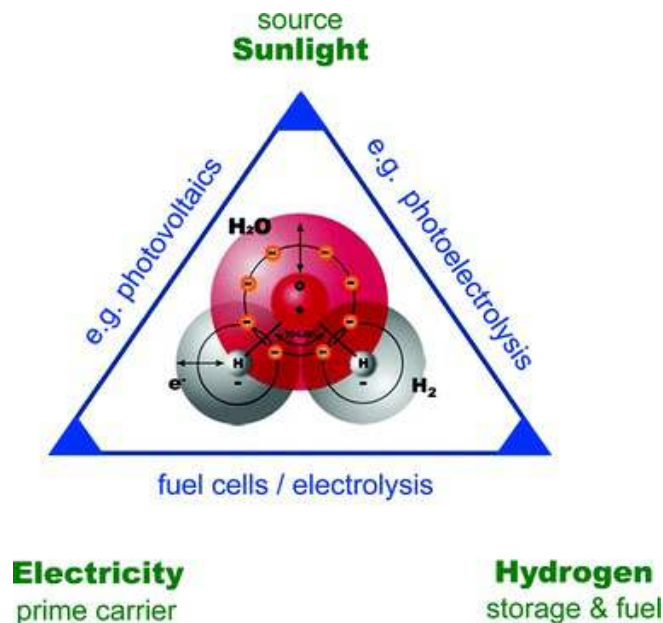
Photoelectrochemical devices...

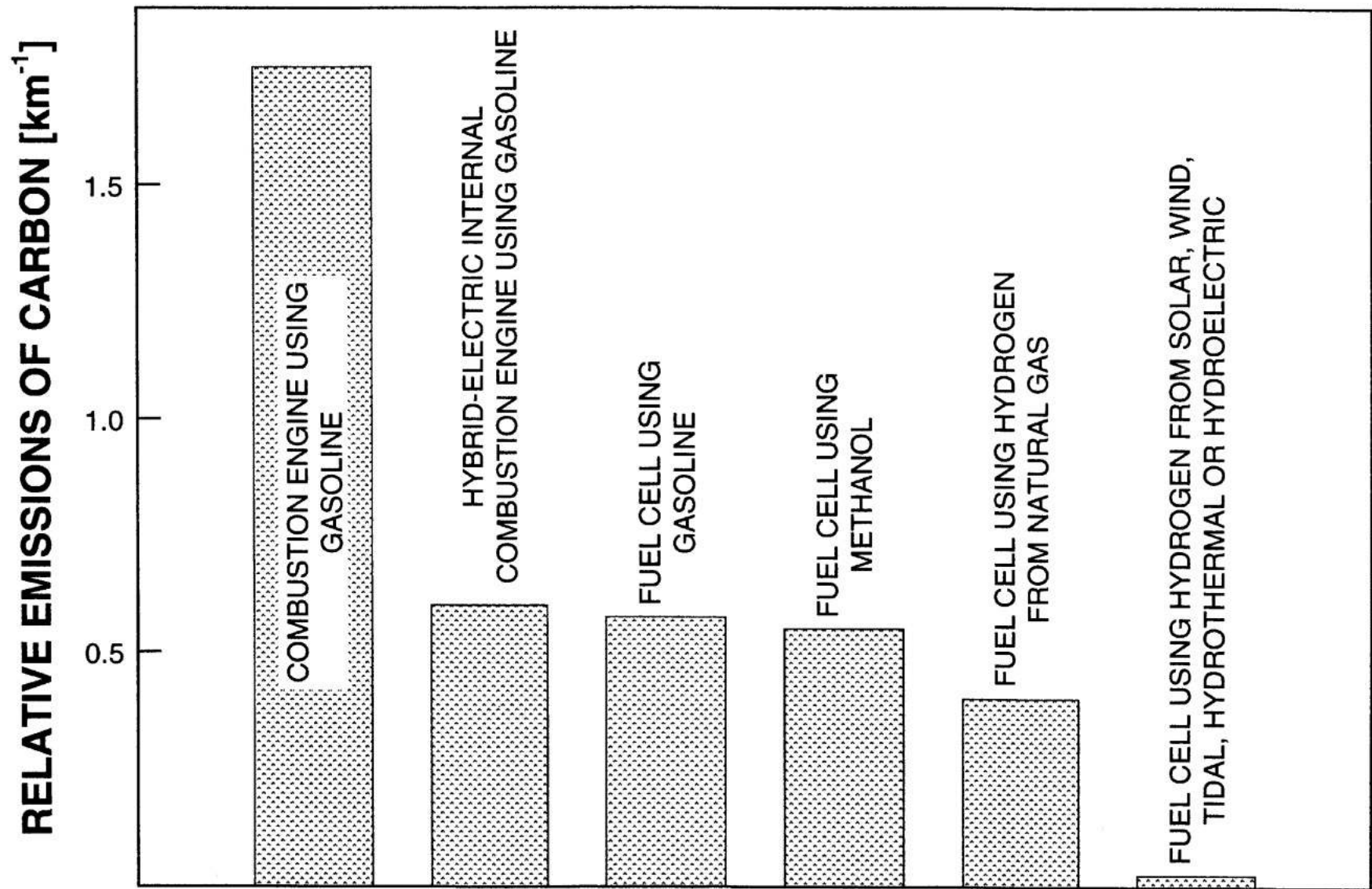
- The main component of the PEC cell is the semiconductor, which converts incident photons to electron–hole pairs.
- At the metal, the electrons reduce water to form hydrogen gas.



Photoelectrochemical devices...

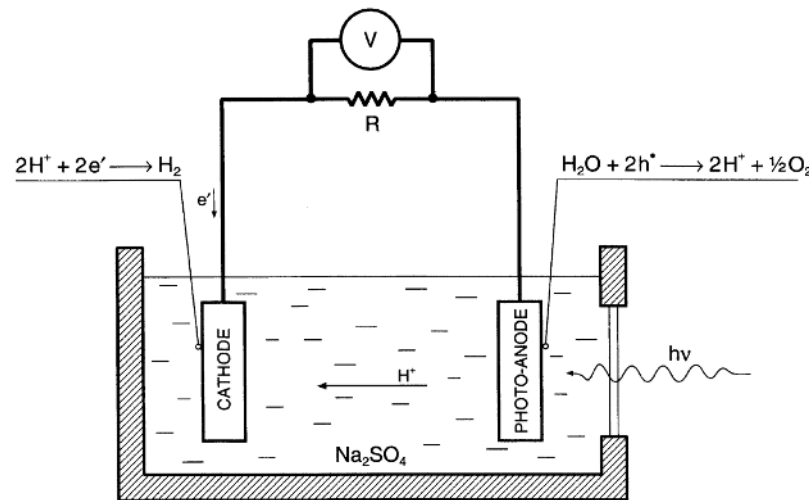
- Hidrogen sebagai bahan bakar tidak menghasilkan polusi.
- Air dan matahari sebagai sumber hidrogen yang melimpah.
- Photoelectrochemical (PEC) water splitting merupakan teknik produksi hidrogen, baik skala kecil maupun besar.
- Honda dan Fujishima (1972) pertama kali investigasi water splitting dengan TiO_2 kristal tunggal sebagai foto-anoda dan Pt sebagai katoda.





Photoelectrochemical devices...

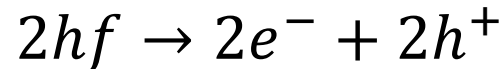
- Listik digunakan untuk elektrolisis air.
- 3 buah kemungkinan susunan PEC
 1. Fotoanoda semikonduktor tipe-n; katoda logam
 2. Fotoanoda semikonduktor tipe-n; Fotokatoda semikonduktor tipe-p
 3. Fotokatoda semikonduktor tipe-p; anoda logam



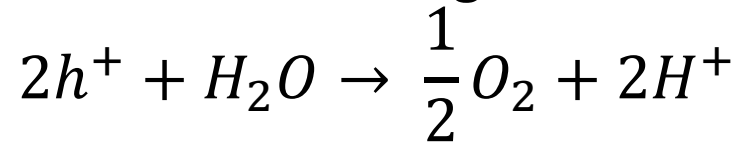
Photoelectrochemical devices...

Mekanisme reaksi:

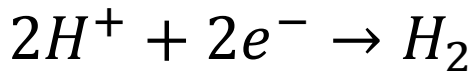
- Fotoelektrolisis menggunakan PEC terdiri dari beberapa tahapan:
- Pancaran foton pada fotoanoda menghasilkan pasangan electron-hole



- Oksidasi air pada fotoanoda dengan hole

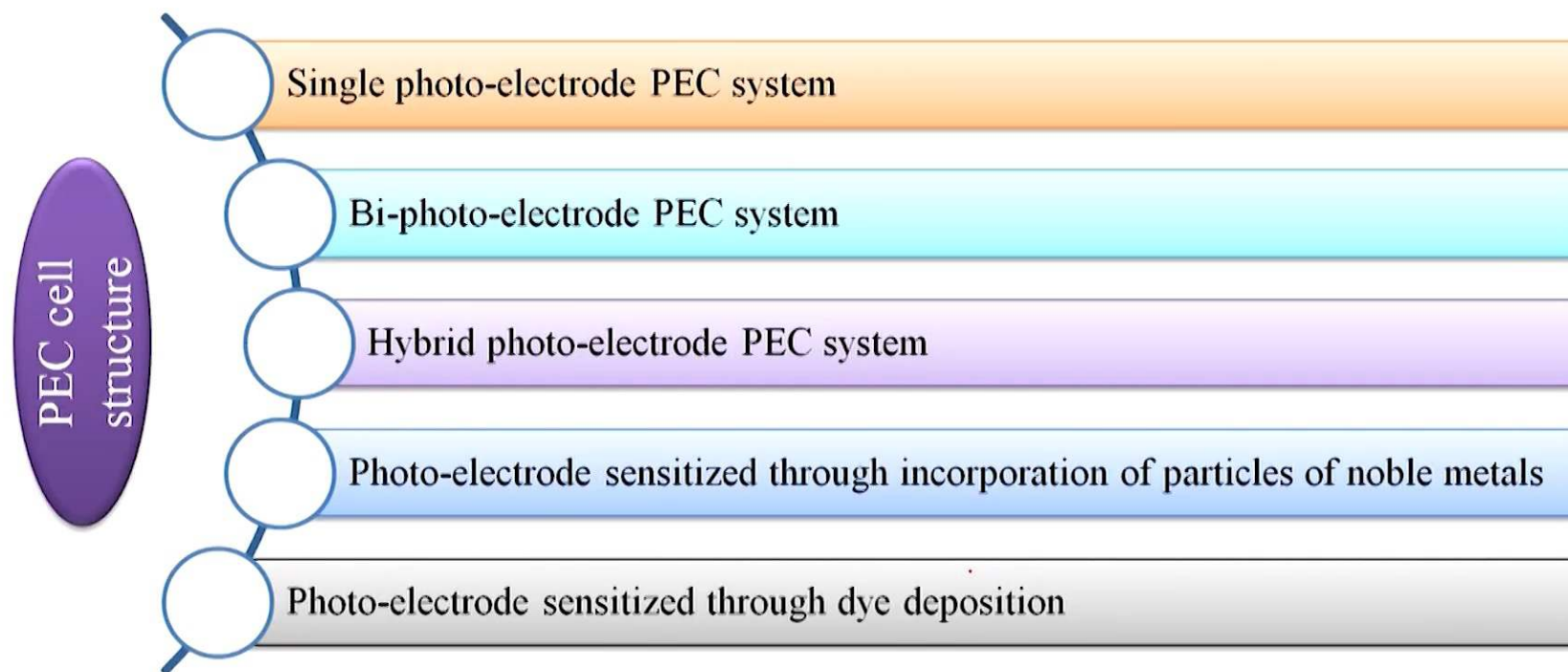


- **Transport ion H^+ dari fotoanoda ke katoda melalui cairan elektrolit**, sedangkan **transport elektron dari fotoanoda ke katoda lewat sirkuit eksternal**.
- Reduksi ion hydrogen oleh electron



Photoelectrochemical devices...

- The PEC cell structures categorized as follows:



1. Single photo-electrode PEC system

- Sama dengan yang dilaporkan oleh Fujishima dan Honda (1972).
- Dilengkapi dengan satu buah fotoelektroda dan logam (tidak sensitive cahaya).

2. Bi-photo-electrode PEC system

- Material Semikonduktor dijadikan sebagai **kedua** fotoelektroda
- Pada kasus ini, tipe-n sebagai fotoanoda sedangkan tipe-p sebagai fotokatoda
- Keunggulan system ini: foto-voltage dibangkitkan pada kedua elektoda yang cukup untuk dekomposisi air tanpa penerapan tegangan.

3. Hybrid photo-electrode (HPE)

- The structure of the HPE, involving the metal contact, Si photovoltaic cell, photo-anode of TiO_2 , and aqueous electrolyte.
- In this structure, only the TiO_2 layer is exposed to the aqueous environment, while the Si solar cell forms a sublayer that is not in contact with the electrolyte.
- The Si solar cell is used to generate a photo-voltage that can be used as an internal electrical bias.
- Consequently, this type of HPE cell is expected to exhibit intrinsic (spontaneous) performance in the absence of an external bias.
- The electrochemical chain incorporated within the HPE is Shown in the 2nd figure.

3. Hybrid photo-electrode (HPE)

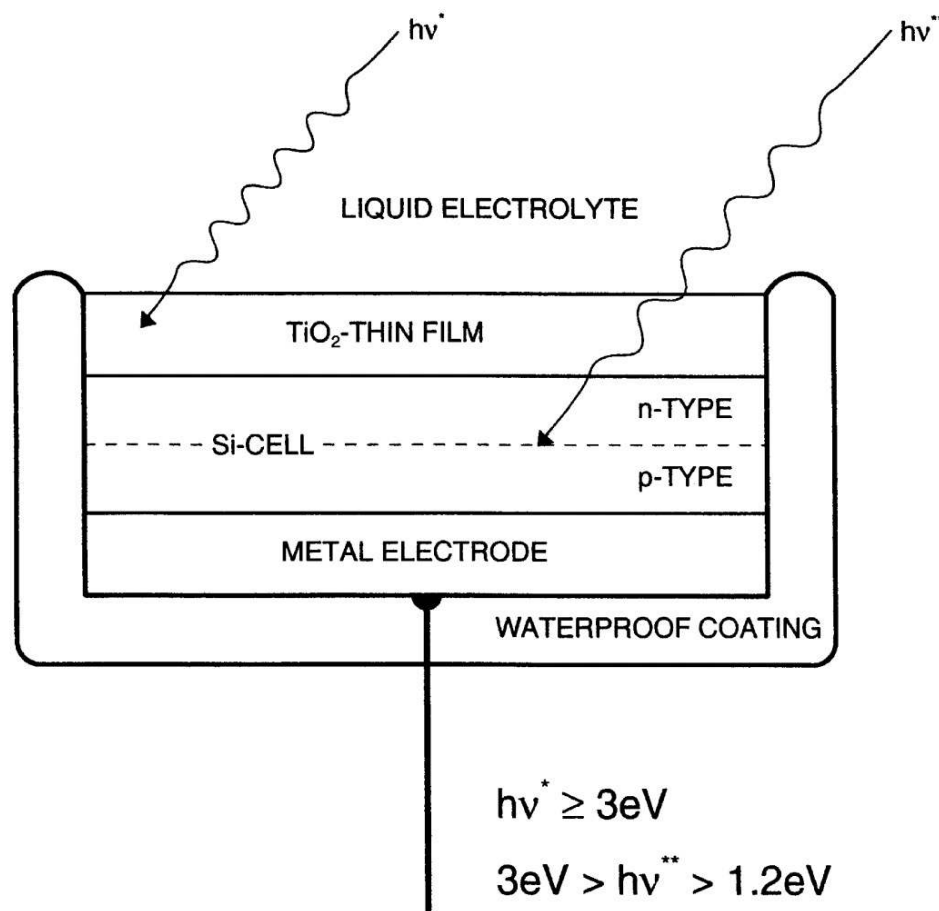


Fig. 18. The hybrid photo-electrode (HPE), involving TiO₂ photo-anode and Si solar cell (according to Morisaki et al. [17]).

3. Hybrid photo-electrode (HPE)

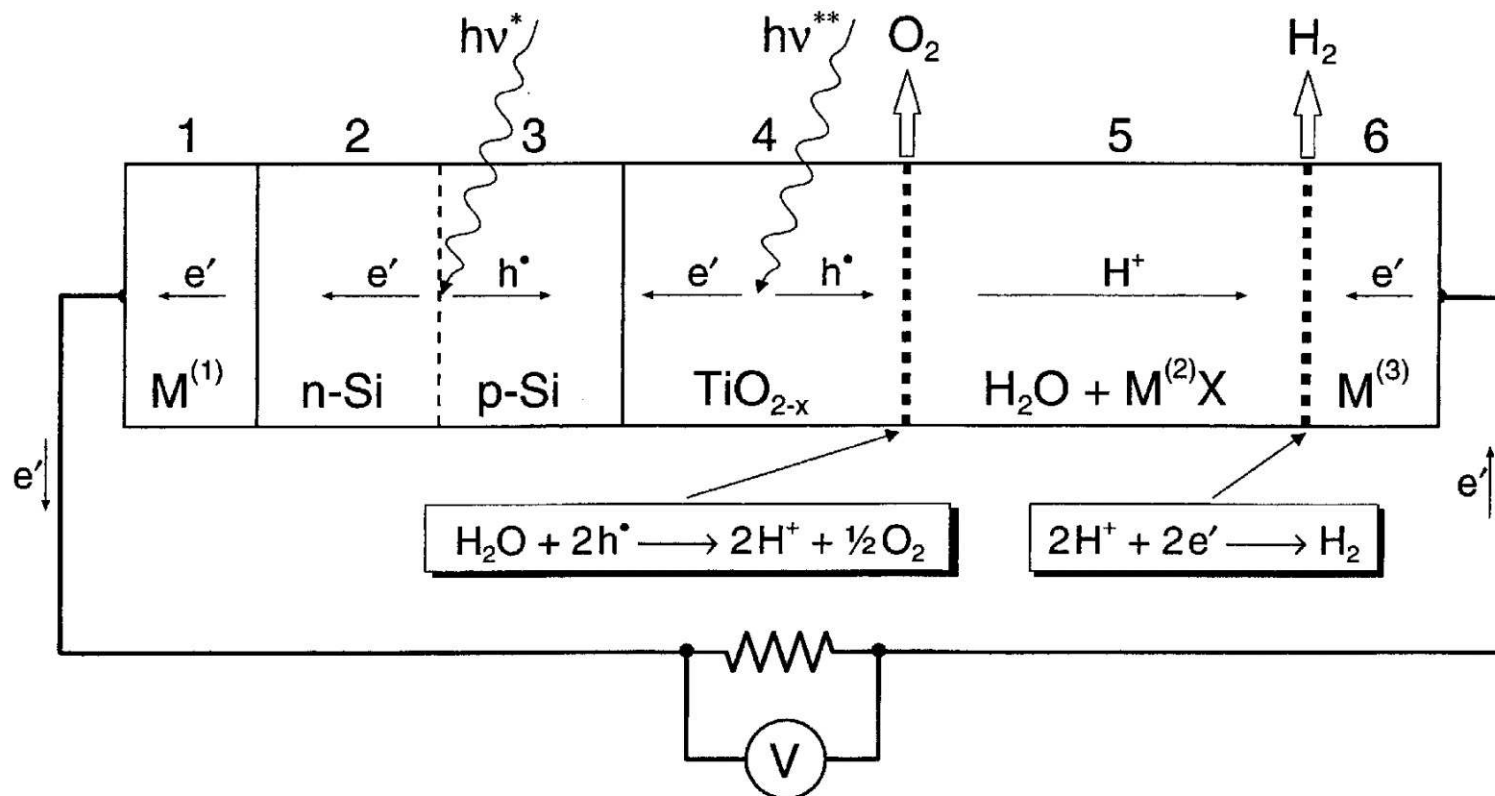


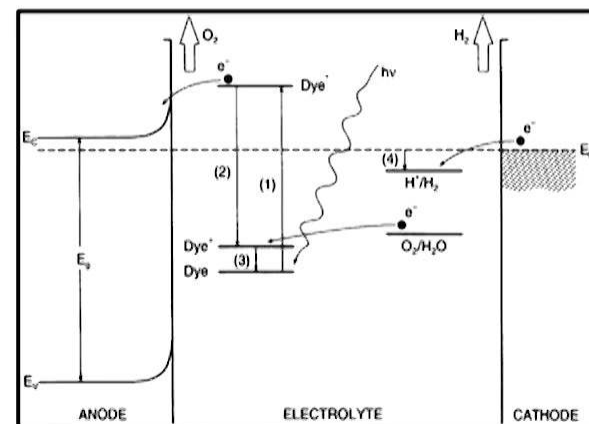
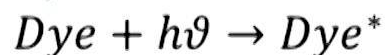
Fig. 19. Electrochemical chain of PEC incorporating HPE shown in Fig. 18.

IV. Photo-electrode sensitized through incorporation of particles of noble metals:

- Photo-anodes (for example, TiO_2 thin films) may be sensitized by **incorporation of small particles of noble metals**, such as **Ag and Pt**, thus forming a dispersed system involving micron-sized particles.

V. Photo-electrode sensitized through dye deposition:

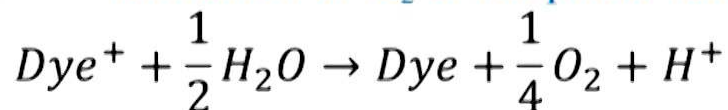
- The conversion efficiency may be increased substantially through **dye-sensitization of photo-electrodes**.
- The **photo-sensitizer**, which is an organic dye, is **physically attached** to the surface of the photo-electrode.
- Light absorption** by the dye, leads to the transformation of the dye molecules from the **ground-state (Dye)** to the **excited-state (Dye*)**:



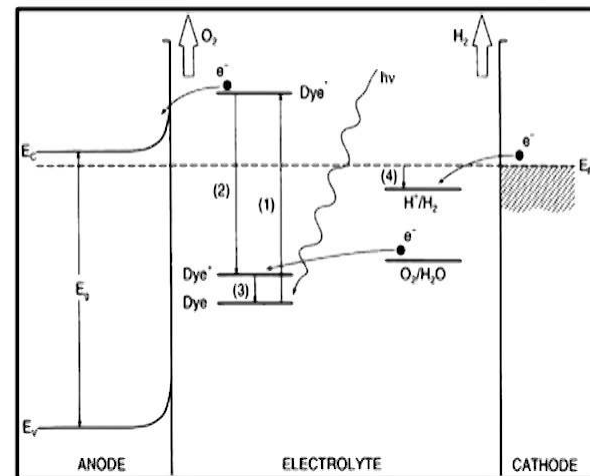
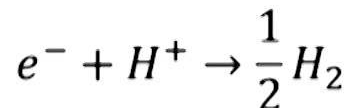
- The transition from the **excited state (Dye^*)** to a **higher oxidation state (Dye^+)** results in the formation of electrons:



- Consequently, the oxidized dye molecule reacts with water, resulting in the **formation of O_2 at the photo-anode**:



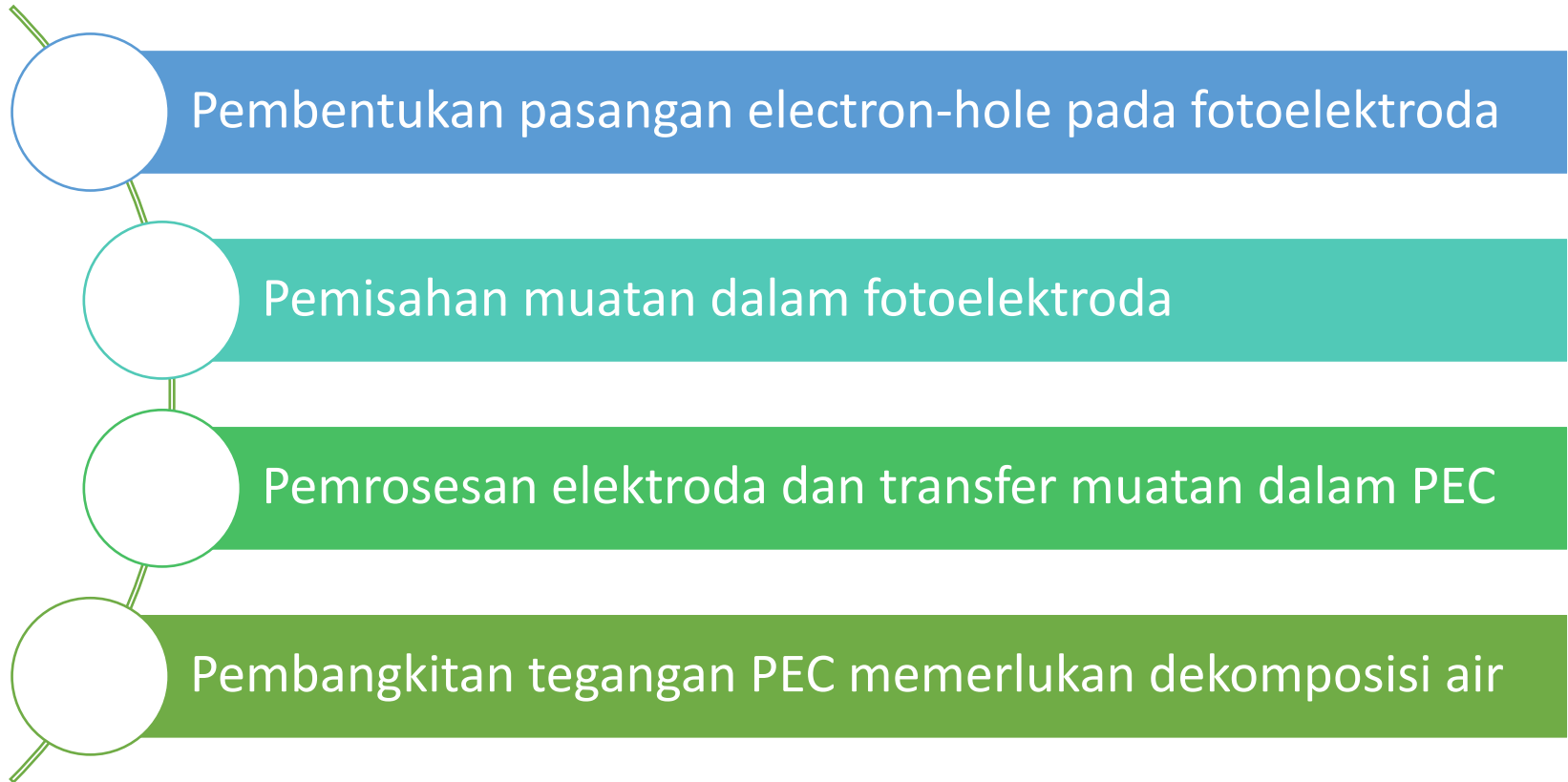
and H_2 at the cathode:



- The dye-sensitized semiconducting **photo-electrode** exhibits two functions,
 - Absorption of light by the dye,
 - Charge transport by the semiconductor.
- Dye sensitization may lead to **sustainable performance of photo-electrodes** only when sensitizers **exhibit a stable performance in aqueous solutions**.

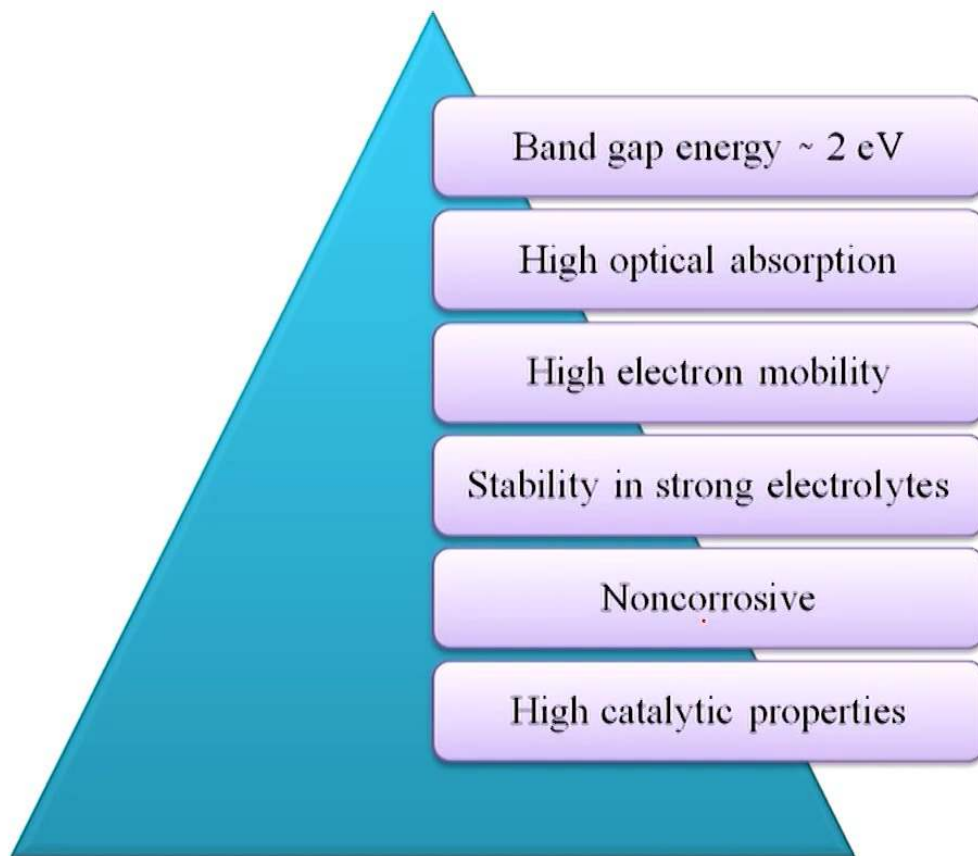
Photoelectrochemical devices...

- **Performa PEC dipengaruhi oleh:**



Photoelectrochemical devices...

- **Kriteria pemilihan material:**



Different nanomaterials used:

C-doped TiO_2 nanocrystalline film, TiO_2 nanotube arrays, ZnO nanostructures, ZnO:Al/ZnO:Ni core/shell nanorods structure, Hematite ($\alpha\text{-Fe}_2\text{O}_3$) nanorods arrays, WO_3 nanowire arrays etc.

Key functional properties of photo-electrodes

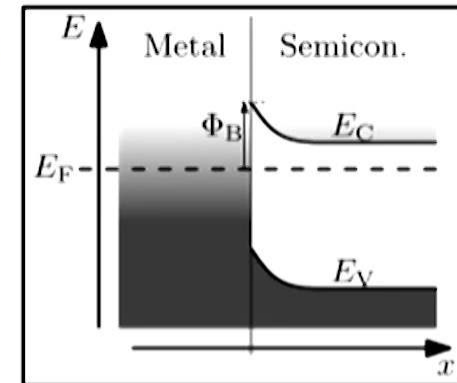
- The materials used for photo-electrodes satisfy several specific functional requirements with respect to **semiconducting and electrochemical properties**.
- Although these properties have been identified, it is **difficult to process materials such that all requirements are satisfied**.

1. Band gap:

- It is an important quantity for materials that are candidates for photo-electrodes.
- A material, which satisfies **high band gap and corrosion resistant**, is not available commercially. Therefore, **there is a need to process such a material**.
- Researchers are working on **hybrid photo-electrode** involving a stack of materials of different band gaps, where each of these absorbs light of different wavelength.

2. Schottky barrier:

- The schottky barrier plays an important role in preventing recombination of the charge formed as a result of photo-ionization.
- An electrical potential barrier across the surface layer can be formed as a result of the following:
 - ✓ Structural deformations within the near-surface layer due to an excess of surface energy;
 - ✓ Segregation-induced chemical potential gradients of aliovalent (having different valence) ions across the surface layer imposed during processing;
 - ✓ Chemical potential gradients of aliovalent ions across the surface layer imposed as a result of surface processing.
- The use of this procedure requires *in situ* monitoring of the surface vs. bulk electrochemical properties during the processing of the electrode materials.

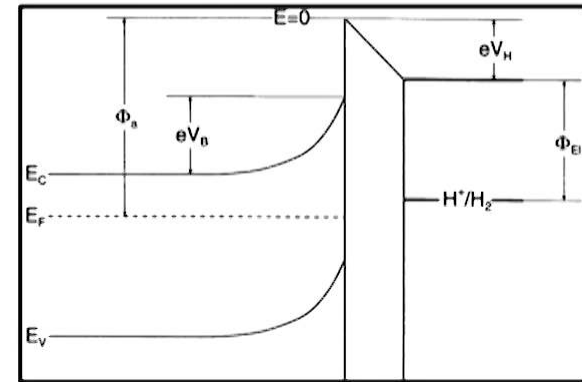


3. Electrical resistance:

- The major sources of energy losses derive from the **ohmic resistances** of the external and internal circuits of the PEC, including:
 - ✓ Electrodes
 - ✓ Electrolyte
 - ✓ Electrical leads (wires)
 - ✓ Electrical connections
 - ✓ Measuring and control equipment
- In order to achieve the **maximum conversion efficiency**, the **electrical resistances of all of these items must be minimized**.

4. Helmholtz potential barrier:

- The charge transfer from the semiconductor to the electrolyte leads to the formation of a surface charge and results in upwards band bending, forming a potential barrier, as shown in Fig.
- This surface charge is compensated by a charge of the opposite sign, which is induced in the electrolyte within a localized layer, known as the Helmholtz layer.
- It is ~ 1 nm thick and is formed of oriented water molecule dipoles and electrolyte ions adsorbed at the electrode surface.
- The height of this potential barrier, known as the Helmholtz barrier, is determined by the nature of the aqueous environment of the electrolyte and the properties of the photo-electrode surface.



5. Corrosion and photo-corrosion resistance:

- An essential requirement for the photo-electrode is **resistance to reactions** at the solid-liquid interface, resulting in degradation of its properties. These reactions include:
 - ✓ Electrochemical corrosion
 - ✓ Photo-corrosion
 - ✓ Dissolution
- **Any form of reactivity results in a change in the chemical composition** and the related properties of the electrode and photo-electrode.
- These processes are particularly **damaging to the properties of the photo-electrode**, where this photo-electrodes are essential for photo-conversion.
- Therefore, is essential for the photo-electrodes to be **resistant to these types of undesired reactions**.

6. Microstructure:

- It is anticipated that future commercial photo-electrodes will be polycrystalline rather than single-crystal.
- So far, little is known of the effect of interfaces, such as surface linear defects caused by grain boundaries, on photo-electrochemical properties of photo-electrodes.
- Therefore, there is an urgent need for the following:
 - ✓ Understanding of the effect of the microstructure of polycrystalline photo-electrodes, specifically that of surface structural defects, on the photo-electrochemical properties.
 - ✓ Development of interface processing technologies leading to the optimization of the energy conversion efficiency of photo-electrodes of polycrystalline materials.

System selection

- The search for high-performance photo-electrodes for solar hydrogen generation includes a wide range of compounds of different compositions, structure, microstructure, and electronic structure.
- While the highest conversion efficiency has been achieved for valence semiconductors, such as GaAs, GaInP, and AlGaAs, these compounds are not promising for practical application because of high costs and poor chemical stability in water.
- On the other hand, metal oxides (MOs) (e.g., TiO_2) exhibit much better stability in water and are less expensive than valence semiconductors.
- The additional advantage of TiO_2 is its high nonstoichiometry and complex defect disorder, which can be used for manipulation with defect-related properties.

Solar energy conversion efficiency

- The overall efficiency of a PEC unit, which is known as the **solar conversion efficiency η_c** , can be defined according to the following formula:

$$\eta_c = \frac{\Delta G_{\text{H}_2\text{O}}^0 R_{\text{H}_2} - V_{\text{bias}} I}{I_r A}$$

Where;

$\Delta G_{\text{H}_2\text{O}}^0$ is the Gibbs free energy of formation for 1 mol of liquid $\text{H}_2\text{O} = 237.141$ (kJ/mol)

R_{H_2} is the rate of hydrogen generation (mol/s)

V_{bias} is the bias voltage applied to the cell (V)

I is the current within the cell (A)

I_r is the incidence of solar irradiance, which depends on geographical location, time, and weather conditions (W/m^2)

A is the irradiated area (m^2)

Advantages:

- ✓ PEC technology depends on solar energy which is a renewable source.
- ✓ It is environmentally safe, with no undesirable byproducts.
- ✓ It may be used on both large and small scales.

Disadvantages:

- ✓ Difficult to get solar energy in the cloudy weathers.
- ✓ It may required complicated optical setups.
- ✓ Device stability, durability, reliability and robustness are major challenges of PEC commercialization.

Topik Bahasan

- Photoelectrochemical water splitting
- Fuel Cells

What is Fuel Cell?

- Divais elektrokimia yang mengkonversi secara langsung dari reaksi elektrokimia menjadi energi listrik.
- Operasinya mirip dengan baterai. Perbedaan: fuel cell menghasilkan listrik, sedangkan baterai menyimpannya.
- Tidak seperti baterai, fuel cell beroperasi terus menerus selagi fuel disuplai.

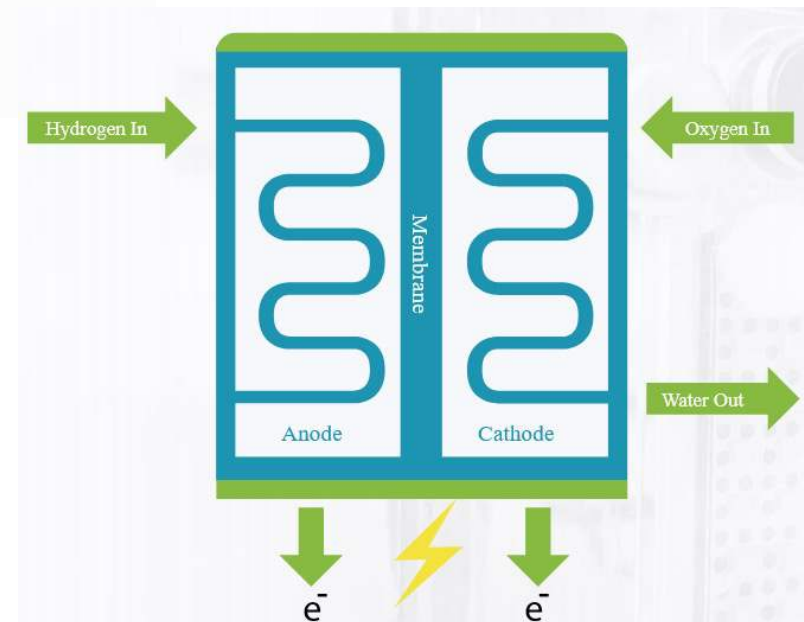
History of Fuel Cells

- 1932: First successful fuel cell device was developed.
- 1959: A practical 5-kW fuel cell system was demonstrated.
- In a fuel cell, hydrogen and oxygen are combined to generate electricity, heat, and water.

How Fuel Cells Work

A fuel cell is an electrochemical energy conversion device – it uses hydrogen and oxygen to generate electricity, heat, and water.

- 1 Hydrogen atoms enter at the anode.
- 2 Atoms are stripped of their electrons in the anode.
- 3 The positively-charged protons pass through the membrane to the cathode and negatively-charged electrons are forced through a circuit, generating electricity.
- 4 After passing through the circuit, the electrons combine with the protons and oxygen from the air to generate the fuel cell's byproducts: water and heat.



Sources of Hydrogen

- **Fossil Fuels**
 - Natural gas
 - Coal
 - Petroleum
- **Biomass**
- **Electrolysis of water - requires cheap, abundant electricity**
 - Electricity from fossil fuels (natural gas and coal)
 - Electricity from renewables (solar, wind, biomass, geothermal, and hydropower)
 - Electricity from nuclear energy
- **Thermochemical production utilizing excess heat from nuclear and concentrated solar**

Sources of Hydrogen

- **Hydrogen does not occur free in nature, and is thus an energy carrier and not a primary energy source; H_2 must be produced from primary energy sources**
- **Natural gas and coal will provide the necessary amount of H_2 to allow the transition to a H_2 infrastructure of the future**
- **Renewables and nuclear will be a significant source of H_2 in the future**

Konstruksi

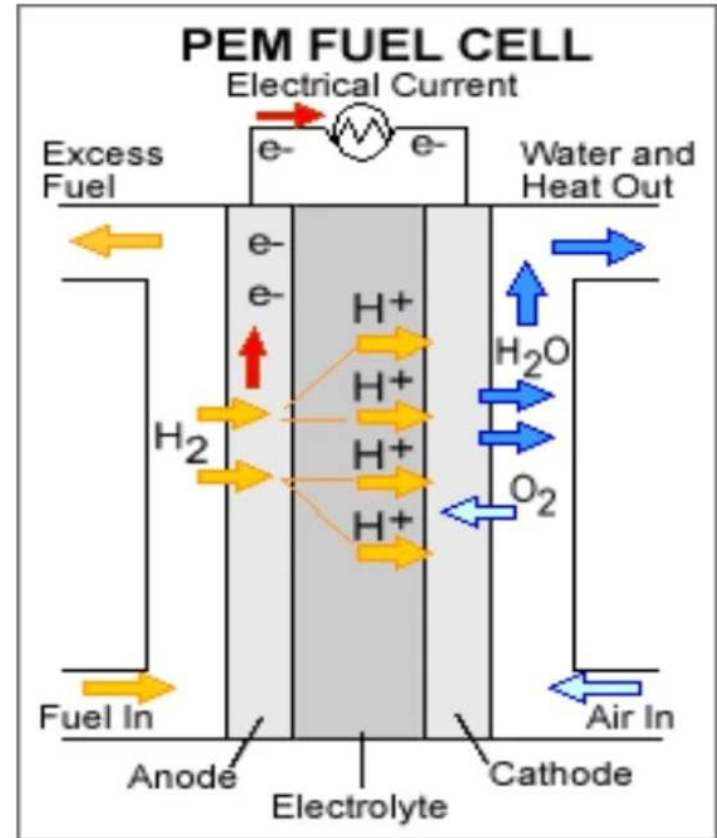
- **Anoda** → menghantarkan electron (dari molekul H_2) agar dapat digunakan oleh sirkuit eksternal
- **Katoda** → menghantarkan electron dari sirkuit eksternal ke kataliss.
- **Elektrolit** → proton exchange membrane
- **Katalis** → memfasilitasi reaksi oksigen dan hidrogen

Types of fuel cells

Fuel Cell type	Electrolyte	Anode gas	Cathode gas	Temp °C	Efficiency %
Proton Ex Membrane (PEM)	Solid polymer membrane	Hydrogen	Pure or Atm Oxygen	75	35 - 60
Alkaline (AFC)	Potassium Hydroxide	Hydrogen	Pure Oxygen	< 80	50 - 70
Direct Methanol (DMFC)	Solid polymer membrane	Methanol soln in Water	Atm Oxygen	75	35 - 40
Phosphoric Acid (PAFC)	Phosphorus	Hydrogen	Atm Oxygen	210	35 - 50
Molten Carbonate (MCFC)	Alkali Carbonate	Hydrogen / Methane	Atm Oxygen	650	40 - 55
Solid Oxide (SOFC)	Ceramic Oxides	Hydrogen / Methane	Atm Oxygen	800 - 1000	45 - 60

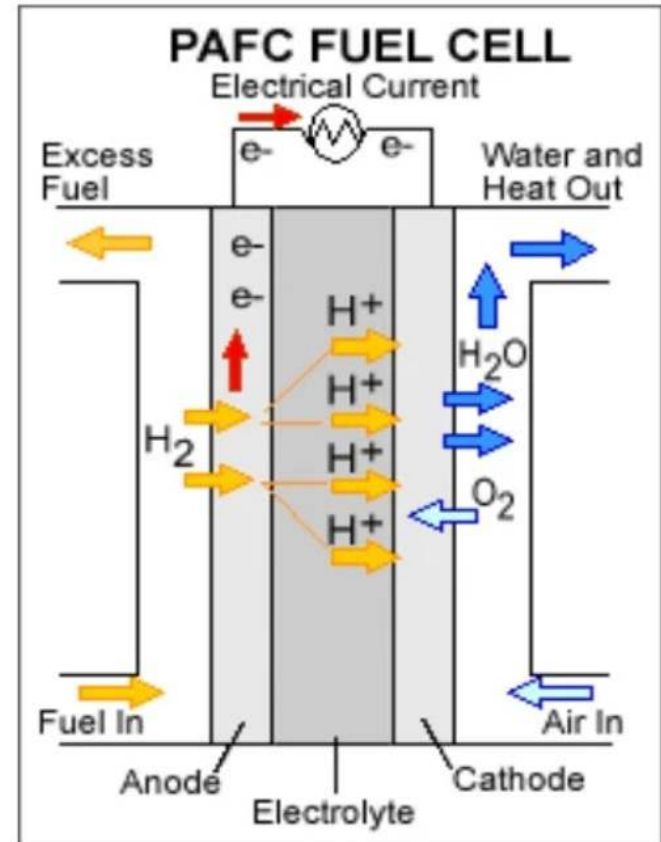
Proton Exchange Membrane

- This is the leading cell type for passenger car application
- Uses a polymer membrane as the electrolyte
- Operates at a relatively low temperature, about 175 degrees
- Sensitive to fuel impurities



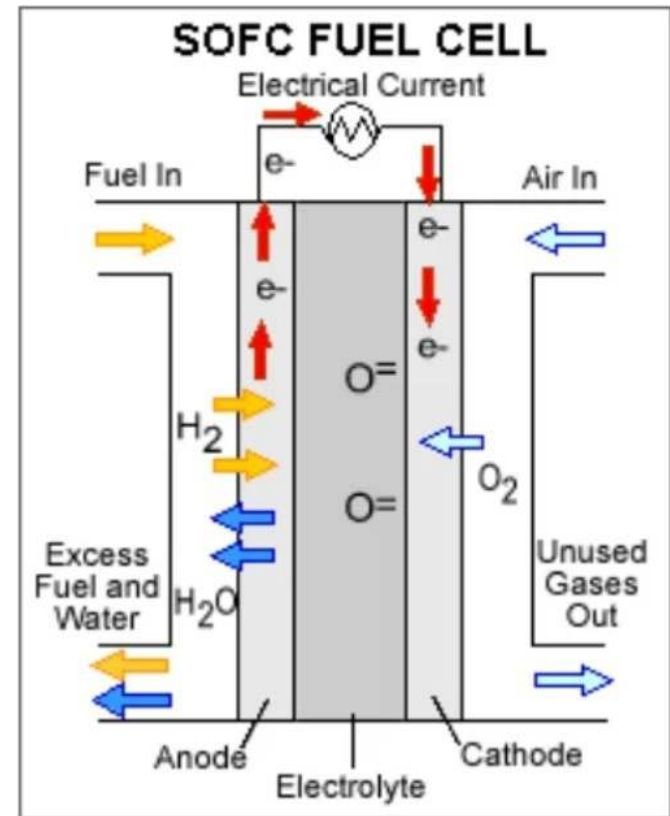
Phosphoric acid

- This is the most commercially developed fuel cell
- It generates electricity at more than 40% efficiency
- Uses liquid phosphoric acid as the electrolyte and operates at about 450 degrees F
- One main advantage is that it can use impure hydrogen as fuel



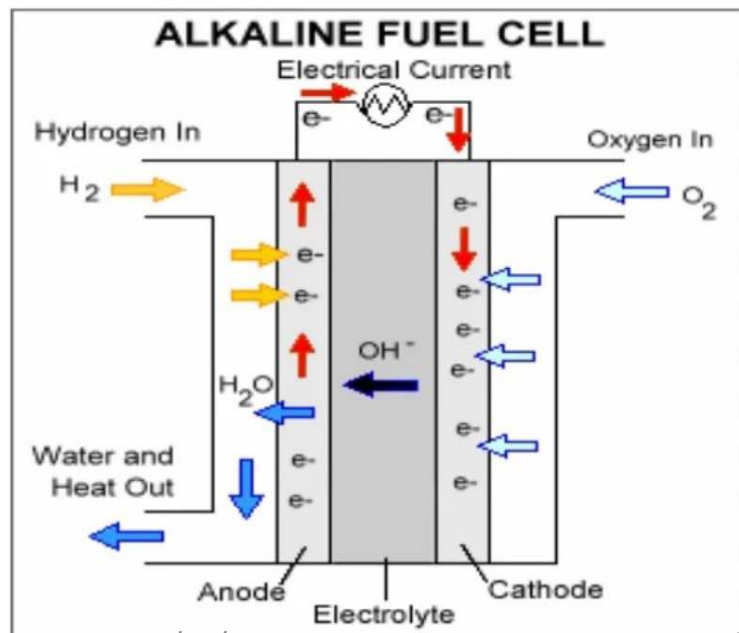
Solid Oxide

- Uses a hard, non-porous ceramic compound as the electrolyte
- Can reach 60% power generating efficiency
- Operates at extremely high temperatures 1800 degrees
- Used mainly for large, high powered applications such as industrial generating stations, mainly because it requires such high temperatures



Alkaline

- Used mainly by military and space programs
- Can reach 70% power generating efficiency, but considered to costly for transportation applications
- Used on the Apollo spacecraft to provide electricity and drinking water
- Uses a solution of potassium hydroxide in water as the electrolyte and operates at 75 -160 degrees
- Can use a variety of non-precious metals as catalyst at the anode and cathode



Pentingnya Hidrogen

- Fuel Cells memerlukan Hidrogen kemurnian tinggi sebagai bahan bakar.
- Peneliti masih mengembangkan teknologi produksi hydrogen yang ekonomis (tantangan) dan ramah lingkungan.

Hydrogen - Today

Production



Steam Reforming
of Petroleum

Use



Space Programme

Storage



Cryogenic Liquid

Hydrogen - Tomorrow

Production



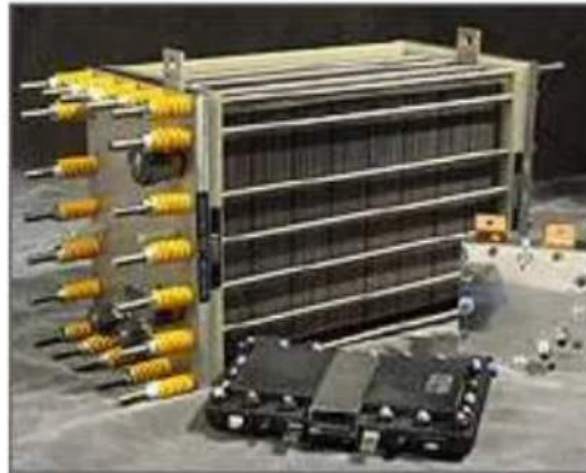
Bio-mass & Electrolysis

Storage



Innovative Tank Designs

Use



Fuel for FUEL CELLS
Electrochemical Energy Conversion

Advantages

- Water is the only discharge (pure H₂)
- Higher efficiency than conventional engines.
- Excellent part load characteristics.
- Zero emission.
- Long operating period between failures.
- No rotating parts in the main hard ware components.
- Negligible noise pollution.

Disadvantages

- CO₂ discharged with methanol reform.
- Little more efficient than alternatives.
- Technology currently expensive.
 - Many design issues still in progress.
- Hydrogen often created using “dirty” energy (e.g., coal).
- Pure hydrogen is difficult to handle.

Applications

- Sumber energi daerah terpencil
- Menyediakan catu daya luar jaringan
- Dapat diaplikasikan pada kendaraan hybrid dan listrik.
- Pabrik pengolahan air limbah dan tempat pembuangan sampah.

Referensi

- Charles C. Sorrell, Sunao Sugihara, Janusz Nowotny - Materials for energy conversion devices - CRC Press (2005)
- Slide Kuliah Prof. Kausik Pal, Photo-electrochemical Production of Hydrogen Using Solar Energy